

Iran-War Supply Chain Risk in Global Chemicals

Coverage date: 2026-03-27 (America/Vancouver)

Focus companies: LyondellBasell (LYB), Dow Inc. (DOW), BASF, SABIC, INEOS, Covestro

Scope: upstream feedstocks and energy, intercontinental logistics corridors, sanctions/trade controls, shipping/insurance market dynamics, and company-level financial sensitivity and mitigations.

Executive summary

A potential (and, as of March 2026, increasingly kinetic) Iran-centered conflict affects chemical companies through three tightly coupled channels: (1) **physical maritime disruption and insurance withdrawal** around the Strait of Hormuz and adjacent Gulf waters; (2) **price shocks in crude/NGL/LNG and refined feedstocks** that rapidly transmit into chemical cost structures; and (3) **sanctions escalation and compliance friction** that can freeze counterparties, vessels, banks, and trade finance even when goods are not Iran-origin. These channels interact: insurance cancellations and risk ratings raise freight and delay cargoes, while price shocks raise working-capital needs; sanctions worsen both by narrowing commercially viable routing, chartering, and payments. ¹

The **Strait of Hormuz is the single highest-consequence single-point failure** for global hydrocarbon-linked chemical cost curves. EIA estimates that in 2024 the Strait averaged **~20 million b/d of oil flows** and that flows through Hormuz in 2024 and 1Q25 represented **more than one-quarter of global seaborne oil trade**; additionally **~one-fifth of global LNG trade** transited Hormuz in 2024 (primarily Qatar). ² Chemical companies do not need to buy Middle East feedstocks directly to be exposed; **global oil/naphtha benchmarks** and LNG-linked power markets transmit disruption into margins across Europe and Asia rapidly.

The **Red Sea-Bab el-Mandeb-Suez system** remains a second critical amplifier. EIA highlights that Persian Gulf exports transiting Suez/SUMED also pass through Bab el-Mandeb and Hormuz, and that closure of Bab el-Mandeb can force diversions around the Cape of Good Hope, increasing time and shipping costs. ³ UNCTAD notes that in 2023 **~22% of global seaborne container trade** passed through the Suez Canal. ⁴ During the 2023–2025 Red Sea crisis, the IMF observed diversions adding **~10+ days** on average—especially damaging for firms with limited inventories. ⁵

Company differentiation (who is most exposed and why):

SABIC is structurally the most directly exposed to Gulf maritime risk because of its Saudi feedstock base, export orientation, and stated monitoring of Bab el-Mandeb attacks with operational responses (alternative routing, inventory updates, storage vessels). ⁶ Dow and LyondellBasell have **material Middle East linkages via JVs** (Dow's Sadara; LYB's Saudi JVs) and strong global shipping footprints; both also explicitly frame feedstock/energy volatility as an earnings driver and use hedging/contract structures to manage it. ⁷ INEOS is highly exposed to disruption in global shipping and feedstock logistics via its large-scale European olefins system and "Trading & Shipping" procurement/transport of ~10 million tonnes/year of gas and liquid feedstocks, plus reliance on maritime access via terminals and long-term contracts; its incremental Middle East exposure is increasing via announced Saudi investments. ⁸ BASF and Covestro

are less directly tied to Gulf production, but are meaningfully exposed through **Europe's naphtha/energy cost curve and intercontinental routing**, and both have documented impacts from Red Sea route bottlenecks and/or explicit energy hedging programs. ⁹

Baseline supply chain and chokepoint mapping

Critical raw materials and where they originate

For the chemical industry in scope, the highest-leverage upstream inputs are:

- **Hydrocarbon feedstocks:** ethane/propane/butane (NGL/LPG), **naphtha**, condensate, and other crude-derived streams that become ethylene/propylene/aromatics (benzene, toluene). Dow explicitly lists purchases of ethane, propane, butane, naphtha, and condensate feedstocks, plus monomers like ethylene and propylene. ¹⁰ LyondellBasell describes ethylene raw materials as NGLs (principally ethane/propane) and crude-based liquids (including naphtha/condensates/gas oils), with ethane comprising ~75% of North American cracker raw materials in 2024. ¹¹
- **Energy inputs:** natural gas, electricity, steam. Many European sites face electricity and gas price exposure and have adopted hedging or procurement strategies for stability (e.g., Covestro hedges part of German electricity and natural gas requirements with an 18-month rolling horizon using derivatives). ¹²
- **Petrochemical intermediates:** Covestro identifies key purchased petrochemical raw materials—phenol, benzene, propylene/propylene oxide, toluene, acetone—collectively **~35% of purchasing value** (2024). ¹³

Maritime corridors, ports, and logistics hubs relevant to an Iran war

Three chokepoints dominate the risk surface for hydrocarbon-linked chemicals and for containerized specialty shipments:

- **Strait of Hormuz (Persian Gulf outlet):** EIA estimates ~20 million b/d oil flows in 2024 and notes Hormuz accounted for >25% of global seaborne oil trade (2024/1Q25) and ~20% of LNG trade in 2024. ¹⁴
- **Bab el-Mandeb (Red Sea gateway):** EIA reports ~6.2 million b/d petroleum flows through Bab el-Mandeb in 2018 and warns closure forces Cape diversions, increasing transit time/cost. ³
- **Suez Canal / SUMED corridor:** EIA describes Suez/SUMED as strategic routes for Persian Gulf crude/products and LNG shipments; total oil flows through Suez+SUMED were ~9% of total seaborne-traded petroleum in 2017, and LNG flows ~8% of global LNG trade. ¹⁵ UNCTAD highlights Suez's containerized trade importance (~22% of global seaborne container trade in 2023). ⁴

Key ports and chemical clusters most implicated in the six-company supply chains:

- **U.S. Gulf Coast (Houston Ship Channel / Freeport corridor):** Port Houston describes the channel as connecting the nation's largest petrochemical complex to the globe. ¹⁶ Dow's major U.S. sites include Freeport and multiple Texas/Louisiana locations. ¹⁷
- **Northwest Europe (Rotterdam, Antwerp-Bruges):** The Port of Rotterdam hosts >45 chemical companies and five refineries and positions itself as a major import/export port for chemicals. ¹⁸

Antwerp-Bruges describes Antwerp as home to the largest integrated chemical cluster in Europe.

19

- **Saudi Arabia Gulf industrial hubs (Jubail / King Fahd Industrial Port):** SABIC reports logistics infrastructure at King Fahad Industrial Port in Jubail related to storage/import/export for key chemicals and intermediates (e.g., the Amiral Storage Project). 20 Dow's partner Aramco similarly frames Jubail east coast siting as strategic access to regional/Asian chemicals markets. 21

Supply chain map (schematic)

```
flowchart LR
    subgraph ME["Persian Gulf producers & chemical hubs (Saudi/UAE/Qatar)"]
        MEF["LPG/NGL, naphtha, polymers\nExport terminals: Jubail/King Fahd Industrial Port,\nJebel Ali, Fujairah, Ras Tanura (illustrative)"]
    end

    subgraph EU["Europe chemical clusters"]
        EU1["Rotterdam (NL)"]
        EU2["Antwerp-Bruges (BE)"]
        EU3["Major integrated sites (e.g., Ludwigshafen, Dormagen/Krefeld, Köln/Lavera)"]
    end

    subgraph US["North America"]
        USGC["U.S. Gulf Coast (Houston Ship Channel / Freeport)\nethane-based and integrated assets"]
    end

    subgraph AS["Asia hubs"]
        AS1["Singapore / China / India\n(major import markets)"]
    end

    MEF -->|Strait of Hormuz| IO["Arabian Sea / Indian Ocean"]
    IO -->|Bab el-Mandeb| RS["Red Sea"]
    RS -->|Suez / SUMED corridor| MED["Mediterranean -> Europe (Rotterdam/Antwerp)"]
    IO --> AS1

    USGC -->|Atlantic| EU1
    USGC -->|Interocean via Panama / Cape (route dependent)| AS1
```

Why this matters for chemicals: feedstocks can be physically constrained by Gulf risk (especially LPG/NGL and LNG), while finished goods and specialty intermediates face freight/lead-time shocks if shippers reroute or insurers restrict cover. EIA explicitly notes pipeline bypass options exist but are limited; it estimates ~2.6 million b/d of Saudi/UAE pipeline capacity might be available to bypass Hormuz in disruption scenarios, which is meaningful but not commensurate with total Hormuz flows. 14

Sanctions and trade restrictions involving Iran

United States (primary constraint for LYB and DOW)

The U.S. Iran sanctions framework materially reduces the plausibility of **direct Iran sourcing** by U.S.-linked chemical companies and their controlled entities, and it increases compliance risk for shipping/insurance/banking interactions.

Under the **Iranian Transactions and Sanctions Regulations (ITSR), 31 CFR Part 560**, U.S. persons are generally prohibited from transactions related to goods/services of Iranian origin or owned/controlled by the Government of Iran, and from export/reexport/supply to Iran (directly or indirectly), absent authorization. ²² OFAC's Iran sanctions program page emphasizes the need to review advisories and program guidance for compliance. ²³

The practical implication for supply chains: even if a company does not buy from Iran, sanctions can block **vessels, intermediaries, terminals, banks, or insurers** involved with Iranian petroleum/petrochemicals. The U.S. Department of State has repeatedly designated entities and vessels for facilitating shipment of Iranian petroleum/petrochemicals (including LPG), signaling ongoing enforcement pressure on logistics networks. ²⁴

European Union and United Kingdom (primary constraint for BASF, Covestro; material for INEOS logistics)

In late 2025, the EU announced reimposition of restrictive measures ("snapback") that had been lifted after the JCPOA, citing the snapback mechanism invoked by the E3 and UN-related developments. ²⁵ Public legal analyses of the snapback describe reintroduced restrictions affecting Iranian crude oil, natural gas, and petrochemical products (referencing Regulation 267/2012 annexes), which would further suppress direct Iran-linked petrochemical trade. ²⁶

The UK also maintains an Iran sanctions regime; UK statutory guidance notes the **Iran (Sanctions) Regulations 2023** entered fully into force in December 2023 and has been updated (as of March 12, 2026). ²⁷

Supply chain consequence: for European manufacturers, the dominant risk is often not direct procurement from Iran, but **counterparty and routing contamination risk**—e.g., chartering or insuring ships that have Iran exposure, transacting via banks or insurers tightening due diligence, and sudden blacklisting of nodes in the shipping ecosystem. ²⁸

Recent incidents and market signals relevant to chemical supply chains

Seizures, attacks, and security advisories

In April 2024, MSC confirmed that the container ship **MSC Aries was seized by Iranian authorities in the Strait of Hormuz** while transiting international waters and that discussions were underway to secure crew release and cargo discharge. ²⁹ An IMO circular letter documenting a communication from Israel similarly

states that IRGC forces seized MSC Aries on 13 April 2024 and forced it toward Iranian territorial waters, with crew taken. ³⁰ Such seizures increase perceived risk for all commercial shipping near Hormuz—raising the probability of rerouting, delays, and war risk premium spikes.

UKMTO continues to publish warnings in the region; for example, UKMTO reported in March 2026 an incident involving a vessel reportedly hit by an unknown projectile near Ras Laffan, Qatar, advising vessels to transit with caution and report suspicious activity. ³¹

Insurance withdrawal and premium repricing

War risk insurance has become a gating variable. In March 2026, industry reporting indicates multiple P&I insurers/clubs issued cancellations or exclusions of war risk cover for vessels operating in Iranian waters and nearby Gulf areas, effective early March. ³² A Gard notice of cancellation explicitly references cancellation of war risks “in Iran and Persian/Arabian Gulf,” forcing insured parties to renegotiate or secure alternative terms. ³³

For the Red Sea corridor, premium volatility has been persistent: the Financial Times reported in mid-2025 that war-risk premiums for Red Sea transit rose to as much as **~1% of ship value**, with a Marsh executive providing indicative cost impacts (hundreds of thousands to ~\$1 million per voyage for a \$100 million vessel). ³⁴ An Insurance Journal report similarly cited Marsh that shipowners paid ~1% of vessel value to pass through the Red Sea during renewed attack periods. ³⁵

Route diversions, transit time, and trade throughput

UNCTAD highlights that the Suez Canal is a major artery for container trade (about 22% of global seaborne container trade in 2023). ⁴ The IMF observed that diversions around the Cape of Good Hope increased delivery times by **~10 days or more on average**, which particularly hurts firms with limited inventories. ⁵

Shipping intelligence has documented sharp drops in Suez traffic during peak risk periods. Lloyd's List Intelligence reported weekly Suez Canal traffic dropping **~34%** over a seven-week period compared with late November (around the onset of major Houthi attacks), illustrating how quickly routing patterns can change. ³⁶ BASF itself noted that in the first half of 2024, EU chemical production rose partly due to “supply bottlenecks on international trade routes through the Red Sea,” indicating real industrial impacts from route disruption. ³⁷

What changed in March 2026

As of late March 2026, reputable reporting indicates a sharp escalation: AP describes Iran formalizing a de facto “toll booth” regime in Hormuz with significant traffic reductions since the onset of conflict, while industry insurance actions (clubs canceling war risk cover) reinforce the commercial severity of Gulf transits. ³⁸

Company assessments

The table below summarizes profiles and key supply chain anchors. A structured, company-by-company analysis follows it.

Company profile and supply chain anchors

Company	Public profile (selected disclosures)	Key products/ segments (high level)	Key feedstocks / energy exposures	Major logistics hubs and routes most relevant to Iran-war risk
Dow (DOW)	Operates 91 manufacturing sites in 30 countries ; major sites include Freeport (TX), multiple Louisiana/Texas assets, and Terneuzen (NL) among others. ¹⁷	Large exposure to ethylene derivatives/ plastics and downstream materials (by segment structure). ¹⁷	Purchases hydrocarbon feedstocks including ethane, propane, butane, naphtha, condensate , plus monomers (ethylene/ propylene) and energy (natural gas, power, steam). Uses “financial and physical hedging programs,” but cannot always pass through costs immediately. ¹⁰	Middle East linkage via Sadara : Dow markets a significant portion of Sadara products; purchases/sells Sadara products; purchases represented ~6% of cost of sales in 2024. ³⁹ Exposed to Gulf routes (Hormuz) for Sadara-related flows and to Red Sea/Suez for Europe–Asia lanes.
LyondellBasell (LYB)	Global petrochemical value chain with U.S./ Europe/Asia reach; explicitly states it obtains a portion of principal raw materials from the Middle East and elsewhere. ⁴⁰	Olefins/ polyolefins, intermediates, and related derivatives (segment detail in 10-K). ⁴¹	European olefins primarily naphtha , with ability to displace up to half with LPG; North American crackers heavily ethane (75% in 2024). Feedstock/energy costs ~75–80% of cost of sales . Uses commodity swaps/options/ futures and contract indexation for risk management. ⁴²	Middle East footprint via Saudi-related ventures/ holdings (and Saudi office/JVs noted publicly). ⁴³ Logistics exposure through Rotterdam/ Antwerp and seaborne naphtha/LPG sourcing into Europe; plus product exports sensitive to Suez/ Red Sea disruptions.

Company	Public profile (selected disclosures)	Key products/ segments (high level)	Key feedstocks / energy exposures	Major logistics hubs and routes most relevant to Iran-war risk
BASF	Highly integrated “Verbund” system; converts feedstocks (mostly naphtha and natural gas) into basic chemicals (ethylene/propylene) and thousands of products. Notes Red Sea trade bottlenecks affected EU chemical conditions in 1H24. ⁴⁴	Broad chemicals portfolio across many value chains (not exhaustively covered here). ⁴⁵	Naphtha/natural gas feedstocks and significant energy demand; central FX hedging via derivatives is disclosed. ⁴⁶	Europe-heavy logistics via Rhine/rail/road plus global marine routes; exposure mainly via European energy/feedstock pricing and Suez-linked flows for EU-Asia trade lanes. ⁴⁷
SABIC	Operates globally; risk reporting explicitly cites Bab el-Mandeb attacks and geopolitical supply chain impacts. ⁴⁸	Major petrochemicals and polymers producer. ⁴⁹	Key raw materials include methane, ethane, propane, butane, naphtha, condensates ; benefits from long-term feedstock agreements with Saudi Aramco . ⁴⁹	High exposure to Hormuz (for Gulf exports) and to Bab el-Mandeb/Suez for Europe-bound flows; SABIC states response plans include alternative routes, inventory updates, and using storage vessels . ⁵⁰

Company	Public profile (selected disclosures)	Key products/ segments (high level)	Key feedstocks / energy exposures	Major logistics hubs and routes most relevant to Iran-war risk
INEOS	Major European olefins/polymer operator; owns large naphtha crackers (Köln, Laverà) and a gas cracker (Rafnes). Rafnes feedstock largely U.S. shale ethane shipped to Europe since 2016; Trading & Shipping procures/transport ~ 10 million tonnes/year of feedstocks. ⁵¹	Olefins & polymers and chemical intermediates (per annual report segment discussion). ⁵²	Primary cracker feedstocks are naphtha and NGLs (ethane/propane/butane) ; relies on maritime access via terminals/storage and long-term contracts, plus third-party shipping managed by in-house marine assurance. ⁵³	Exposure is strongest to global shipping disruption (Red Sea/Suez) and to global naphtha/energy benchmarks; incremental Middle East exposure via announced Saudi investment (acrylonitrile project). ⁵⁴
Covestro	Specialty polymer group with global manufacturing network; identifies major petrochemical raw materials (phenol/benzene/propylene oxide/toluene/acetone) as 35% of purchasing value. ¹³	Polycarbonates, precursors for coatings/adhesives, MDI/TDI, TPU, films, elastomers, etc. ⁵⁵	Procurement uses long-term supply agreements; explicitly hedges part of German electricity and natural gas needs with an 18-month rolling horizon using derivatives. ¹²	Exposure mainly via European feedstock/energy costs and EU-Asia shipping; less direct Gulf physical exposure than SABIC/Dow/LYB but still vulnerable to Suez/Red Sea freight spikes. ⁵⁶

Direct vs. indirect Iran sourcing (all companies)

Across the six companies, **direct sourcing from Iran is unlikely to be material** absent specific authorizations because of U.S./EU/UK sanctions constraints that prohibit or heavily restrict Iran-linked trade, especially in energy and petrochemicals. For U.S. persons, ITSR broadly prohibits transactions involving Iranian-origin goods/services and export/reexport/supply to Iran without authorization. ²² The EU's 2025 snapback announcement indicates tightened restrictive measures, and UK guidance reflects a standing Iran sanctions regime. ⁵⁷

The more durable risk is **indirect exposure**: benchmark price transmission (oil/naphtha/LNG), shipping interruptions and insurance, and compliance disruption (counterparty screening, blocked vessels, denied coverage, and payment delays). ⁵⁸

Company-by-company risk analysis against the requested attributes

Dow (DOW)

Dow's disclosed feedstock slate (ethane/propane/butane/naphtha/condensate) and reliance on purchased energy means a Hormuz-driven hydrocarbon shock can affect input costs broadly, though Dow also expects U.S. NGL supplies to remain abundant and uses flexibility and hedging programs to mitigate increases. ¹⁰

The distinctive Iran-war sensitivity is **Sadara**: Dow disclosed it markets much of Sadara's output outside the Middle East and purchases/sells Sadara products; purchases represented **~6% of cost of sales** in 2024. ³⁹

A Gulf disruption that reduces Hormuz transits or raises Gulf-area war risk insurance can affect both physical availability and delivered cost of Sadara-related product flows (even if production continues), and can also increase working capital through longer or more circuitous routings.

Dow's ability to pass through cost increases is explicitly limited by market conditions and timing ("not always able to immediately raise selling prices"). ¹⁰ Therefore short-term spikes in feedstock/freight costs (days to weeks) are most likely to compress margins before pricing resets. Mitigation levers include feedstock flexibility, financial/physical hedging, rebalancing production among advantaged sites, and contracting adjustments—supported by Dow's broad global manufacturing footprint (91 sites in 30 countries). ⁵⁹

LyondellBasell (LYB)

LYB's disclosed cost structure makes it acutely sensitive to feedstock/energy: it states feedstock and energy costs are **~75-80% of cost of sales**. ⁴¹ This is a core reason an Iran-war scenario matters even without direct Iran procurement. LYB's Europe exposure is particularly naphtha-linked; it states naphtha is the primary raw material in European olefins facilities, with flexibility to displace up to half with LPG. ⁶⁰ In a Hormuz disruption, global crude/naphtha benchmarks can move sharply; additionally, LPG availability and price can shift as exporters and buyers reroute. In North America, LYB reports ethane comprised **~75%** of North American cracker raw materials in 2024, which can be structurally more insulated from oil-linked naphtha shocks (though not immune to broad energy volatility). ⁶¹

LYB explicitly states it obtains a portion of principal raw materials from the **Middle East**, and it highlights political instability in those regions as potentially increasing prices and reducing supply. ⁴⁰ That creates a more direct channel than BASF/Covestro for some commodity feedstocks and intermediates. LYB also discloses commodity derivatives usage (swaps/options/futures) and contract indexation, with a quantified sensitivity: a 10% commodity price change would change commodity derivative fair value by about **\$45 million** (as of Dec. 31, 2024). ⁶²

Mitigation options: maximize feedstock flexibility (naphtha↔LPG substitution in Europe), increase geographic production balancing, expand customer contract pass-through where feasible, and pre-position feedstocks/products where war-risk rerouting introduces 10+ day lead time extension (as observed by IMF during Red Sea diversions). ⁶³

BASF

BASF's exposure is primarily **Europe's energy and naphtha cost curve** and intercontinental logistics. BASF explains its Verbund converts feedstocks—mostly **naphtha and natural gas**—into basic chemicals like

ethylene and propylene that feed downstream chains. ⁴⁵ While BASF does not position itself as Middle East feedstock dependent in the same direct way as SABIC, it is exposed to any sustained global crude/naphtha price shock. BASF also documents that Red Sea disruptions created “supply bottlenecks on international trade routes through the Red Sea” affecting market conditions in 1H24. ³⁷ That is direct evidence of sensitivity to route insecurity even without Hormuz closure.

Risk mitigations are more structural and operational than tactical stockpiling: integrated Verbund site optimization, rerouting shipments (where feasible), and financial risk management (BASF discloses centralized FX hedging using derivatives). ⁶⁴ In a prolonged conflict, Europe’s competitiveness versus gas-advantaged regions can weaken, and BASF may face margin pressure or demand destruction. The company’s major vulnerability is less about Iranian suppliers and more about **macro feedstock/energy and freight regimes**.

SABIC

SABIC is the most explicitly “Middle East-centered” profile in this set. SABIC states it relies on gas and crude oil-based petrochemical raw materials including methane, ethane, propane, butane, naphtha, and condensates, and that in Saudi Arabia it benefits from **long-term feedstock agreements with Saudi Aramco**. ⁴⁹ This implies strong upstream security under normal conditions, but it does not eliminate Gulf transit risk: a Hormuz disruption affects exports and the delivered economics of feedstocks/products.

SABIC’s own risk disclosure highlights 2024 geopolitical disruptions: it explicitly points to “attacks in the Bab-el-Mandeb Strait” and says its response plans include **updating inventory, considering alternative shipping routes, and utilizing storage vessels**. ⁴⁸ This is a rare, direct acknowledgment of maritime security translating into supply chain operating measures.

Mitigation levers are partially structural (Saudi Aramco feedstock agreements; port storage infrastructure such as projects associated with King Fahad Industrial Port in Jubail) and partially tactical (inventory/stowage and routing). ⁶⁵ In a wide regional escalation, SABIC may face both transit disruption and demand shocks in Asia/Europe; but it may also gain pricing power in regions facing supply shortages, depending on its ability to physically deliver.

INEOS

INEOS stands out for its **explicit logistics architecture**. It owns major crackers (Köln naphtha crackers; Rafnes gas cracker; Lavera naphtha cracker) and describes a multi-year pivot to U.S. shale ethane supply for Rafnes, supported by long-term supply and infrastructure agreements and dedicated “Dragon” vessels transporting feedstocks to Europe. ⁶⁶ Within Olefins & Polymers Europe, INEOS states Trading & Shipping is responsible for procurement, trading, and transportation of **~10 million tonnes/year** of feedstocks including ethane, propane, butane, and naphtha, and notes reliance on shipping channels is mitigated via ownership/stakes in terminals and storage or long-term access contracts. ⁵³

This means an Iran-war scenario affects INEOS primarily through **global shipping market dislocation and war-risk geography**: if the Red Sea/Suez is avoided, Europe-Asia lanes lengthen (10 days to two weeks in some analyses), tightening vessel availability and potentially cascading into higher costs or reduced reliability for INEOS’s shipping-managed supply chain. ⁶⁷ It also has incremental strategic exposure to the Middle East via announced Saudi investments (e.g., an acrylonitrile plant plan outlined by INEOS). ⁶⁸

Mitigations include long-term feedstock logistics contracts (already core to its model), diversification of routing, and potentially inventory buffering at terminals—consistent with INEOS’s emphasis on terminals/storage stakes. ⁶⁹

Covestro

Covestro’s highest Iran-war sensitivity is indirect: (1) it purchases petrochemical raw materials (phenol/benzene/propylene/PO/toluene/acetone) that are benchmark-linked to crude/naphtha, and these account for **~35% of purchasing value**; (2) its European operations face energy price risk; and (3) intercontinental logistics may be disrupted by Suez/Red Sea rerouting. ⁷⁰

Covestro discloses active mitigation: long-term supply agreements and supplier management, plus explicit energy hedging—since fiscal 2024 it hedges a portion of German electricity and natural gas needs with an **18-month rolling horizon** using derivatives, and it also hedges the natural gas component embedded in steam pricing under relevant contractual arrangements. ¹² This is directly relevant if an Iran conflict tightens LNG markets and European gas prices respond.

For Covestro, disruption scenarios most plausibly show up as short-term margin compression due to input and freight spikes and as working-capital increases from longer transit times and higher inventory requirements (consistent with IMF observations about inventory constraints under Red Sea diversions). ⁵

Scenario analysis and mitigation playbook

Scenario framing (what “Iran war” means operationally)

This report uses three scenarios with distinct supply chain signatures:

Short-term disruption (days to weeks): episodic missile/drone events, seizures, and insurer premium spikes; intermittent convoying and localized port slowdowns; partial rerouting and higher war-risk premiums; oil/naphtha price spikes with uncertain duration. March 2026 insurance cancellations (war risk cover exclusions) are consistent with this regime becoming commercially binding. ⁷¹

Prolonged conflict (months): persistent avoidance of Hormuz and/or sustained “managed transit” regimes; durable shift to Cape of Good Hope routings; carrier schedule unreliability; sustained high war-risk and freight rates; elevated working capital requirements; demand destruction in consumer and industrial end markets due to energy-driven inflation.

Escalation to wider regional war (multi-theater): broad attacks affecting Gulf and Red Sea simultaneously; large-scale interruption of Hormuz transits and Gulf-area port operations; widespread insurance withdrawal; sanctions intensification; severe crude/LNG shock and global recession risk.

Chokepoint exposure metrics (context chart)

The values below are **indicator-level** (not a complete model) but show why these chokepoints dominate chemical supply chain risk.

```

xychart-beta
  title "Chokepoint oil-trade exposure indicators (selected EIA statistics)"
  x-axis ["Strait of Hormuz (2024/1Q25)", "Bab el-Mandeb (2017 basis)",
"Suez+SUMED (2017 basis)"]
  y-axis "Approx. share of seaborne petroleum trade (%)" 0 --> 30
  bar [25, 9, 9]

```

EIA states Hormuz flows in 2024/1Q25 made up **more than one-quarter** of global seaborne oil trade, while Bab el-Mandeb and Suez+SUMED each accounted for **~9%** of total seaborne-traded petroleum in 2017 (with Bab el-Mandeb also described as a gateway for Gulf exports to Suez/SUMED). ⁷²

Company mitigation options (what can be done, and by whom)

Cross-industry mitigations (applicable to all six):

Operationally, the highest-ROI mitigations are to (1) **increase routing optionality** (contracts with carriers that can support Cape diversions; alternative load/discharge ports; transshipment planning), (2) **increase buffer inventories** of critical intermediates where safety/regulatory constraints allow (especially for specialty inputs with limited substitutes), and (3) **tighten sanctions and vessel due diligence** to reduce blocked-transaction risk amplifications (screening vessel histories, terminals, insurers, and banks). These are consistent with observed impacts in Red Sea disruptions—longer transit times (10+ days) and higher insurance premiums drive the need for higher inventory and revised logistics planning. ⁷³

Company-specific mitigation emphasis: - **SABIC:** double down on alternative routing and storage strategies already cited (inventory updates, storage vessels), and maximize Saudi export optionality (Red Sea ports where feasible) given EIA's discussion of Saudi pipeline bypass capacity to Yanbu when Bab el-Mandeb flow risks rise. ⁷⁴

- **Dow:** diversify marketing and sourcing channels for Sadara-linked products; build contractual flexibility for Gulf disruptions; use site balancing across its global footprint and maintain hedging discipline recognizing pass-through delay risk. ⁷⁵

- **LYB:** maximize Europe feedstock flexibility (naphtha↔LPG) and actively manage commodity derivatives and indexed contracts to reduce cost spikes; pre-position inventories for critical customers/regions under longer shipping lead times. ⁷⁶

- **INEOS:** leverage existing terminal/storage stakes and long-term logistics contracts; ensure ship/terminal insurance arrangements remain viable under changing war-risk clauses (BIMCO's war risk clause updates reflect the contractual importance of allocating premium and rerouting rights). ⁷⁷

- **BASF / Covestro:** focus on European energy and feedstock risk management, including hedging or long-term supply contracting (Covestro's explicit 18-month hedging horizon is a leading indicator of best practice for crisis buffering). ⁷⁸

Comparative tables and timeline

Comparative risk matrix (requested attributes)

Scores are qualitative (Low/Medium/High), grounded in disclosed footprints and the chokepoint/sanctions environment; they are intended for **comparative prioritization**, not precision forecasting.

Attribute	Dow	LYB	BASF	SABIC	INEOS	Covestro
Direct sourcing from Iran (expected)	Low (sanctions constraints for U.S. persons) ²²	Low (sanctions constraints; compliance risk noted) ²²	Low (EU/UK regime; snapback) ⁵⁷	Low-Medium (not asserted; but regional proximity heightens inadvertent exposure risk) ⁴⁸	Low (EU/UK regime; shipping counterparties risk) ⁷⁹	Low (EU/UK regime; procurement controls) ⁸⁰
Indirect exposure to Iran conflict via oil/LNG price shocks	High (broad hydrocarbon feedstocks; pass-through delay) ⁸¹	High (feedstock/energy 75–80% of COGS) ⁸²	High (naphtha/natural-gas-based Verbund) ⁸³	High (hydrocarbon feedstock base; export economics) ⁸⁴	Medium-High (naphtha/NGL dependence; European benchmark exposure) ⁸⁵	Medium-High (petrochemical raw materials 35% of purchasing value; energy hedging mitigates) ⁸⁶
Exposure to Hormuz / Gulf war-risk disruption	Medium-High (Sadara-linked flows) ⁸⁷	Medium (Middle East raw materials + Saudi footprint) ⁸⁸	Low-Medium (mostly indirect via markets) ¹⁴	High (Gulf export corridor; ERM tracks conflict) ⁷⁴	Low-Medium (less Gulf-centric today; rising via Saudi investments) ⁸⁹	Low-Medium (mostly indirect) ¹⁴
Exposure to Suez/ Red Sea/ Bab el-Mandeb disruption	Medium (global trade lanes) ⁹⁰	Medium-High (Europe feedstock imports and exports) ⁶³	Medium-High (documented Red Sea bottleneck impacts) ⁹¹	High (explicitly monitors Bab el-Mandeb attacks; routing/inventory responses) ⁵⁰	High (shipping-centric feedstock procurement; war-risk contract sensitivity) ⁷⁷	Medium (Europe-Asia logistics; energy input exposure) ⁹²

Attribute	Dow	LYB	BASF	SABIC	INEOS	Covestro
Inventory / stockpile posture (public signals)	Medium (global network; but continuous-process constraints) ¹⁷	Medium (large commodity system; uses derivatives and indexed contracts) ⁹³	Medium (Verbund integration; rerouting more than stockpiling) ⁴⁴	Medium-High (explicit inventory updates + storage vessels) ⁴⁸	Medium-High (terminal/storage stakes; long-term access contracts) ⁶⁹	Medium (procurement + energy hedging; not a "stockpile" business) ⁹⁴
Contract / hedging strategies (disclosed)	Hedging programs + feedstock flexibility; pass-through limits ¹⁰	Commodity swaps/options/futures; indexed contracts; quantified sensitivity ⁹³	Central FX hedging using derivatives ⁶⁴	ERM + alternative routes/inventory response (hedging not detailed here) ⁹⁵	Terminal/storage + long-term shipping and feedstock contracts ⁵³	Long-term supply + explicit electricity/gas hedging (18-month horizon) ¹²
Financial sensitivity (evidence level)	Medium-High (COGS sensitive to feedstock/energy; Sadara purchases 6% of COGS) ⁹⁶	High (feedstock/energy 75-80% of COGS) ⁴¹	Medium-High (naphtha/natural gas central to production economics) ⁴⁵	Medium-High (geopolitical disruptions cited as raising shipping/energy costs) ⁹⁵	Medium-High (margin and logistics drivers tied to feedstock mix and shipping) ⁵³	Medium (raw material share quantified; explicit energy hedging helps) ⁹⁷

Timeline of recent Iran-related shipping events (mermaid)

timeline

2023-11-19 : Major Red Sea insecurity begins (Galaxy Leader hijacking referenced in shipping intelligence coverage)

2023-12-18 : Maritime protection coalition begins (Operation Prosperity Guardian era)

2024-01-14 : Suez Canal transits drop sharply vs late Nov (shipping intelligence data)

2024-03-07 : IMF notes Cape diversions add ~10+ days, stressing low-inventory supply chains

2024-04-13 : IRGC seizes MSC Aries in Strait of Hormuz (confirmed by MSC; documented in IMO circular communication)

2024-10 : UNCTAD/Suez briefing highlights rising insurance premiums and longer sail times during Red Sea crisis

2025-07 : Red Sea war-risk insurance costs spike again (reports cite up to ~1% of ship value)
 2025-09-29 : EU announces snapback reimposing restrictive measures on Iran
 2025-10 : U.S. sanctions target networks trading Iranian petroleum/ petrochemicals (incl. LPG logistics)
 2026-03-05 : P&I clubs/insurers cancel war-risk cover in Iran/Gulf areas; voyages reprice and reroute
 2026-03-18 : UKMT0 reports projectile incident near Ras Laffan, Qatar; cautions vessels
 2026-03-27 : AP: Iran formalizes Hormuz “toll booth” regime amid conflict; traffic sharply reduced

Source grounding for the timeline: Suez traffic drop reporting (Lloyd’s List) ³⁶ ; IMF diversion timing ⁵ ; MSC Aries seizure confirmation (MSC) and IMO circular communication ⁹⁸ ; UNCTAD/Suez crisis briefing metrics ⁹⁹ ; Red Sea war-risk premium spike ¹⁰⁰ ; EU snapback ²⁵ ; U.S. petroleum/petrochemical sanctions action ²⁴ ; March 2026 war-risk cancellations (Insurance Journal + Gard notice) ⁷¹ ; UKMT0 March 2026 warning ³¹ ; AP March 2026 Hormuz controls ¹⁰¹

¹ ²⁸ ³² ⁷¹ <https://www.insurancejournal.com/news/international/2026/03/02/860022.htm>
<https://www.insurancejournal.com/news/international/2026/03/02/860022.htm>

² ¹⁴ ⁵⁸ ⁷² <https://www.eia.gov/todayinenergy/detail.php?id=65504>
<https://www.eia.gov/todayinenergy/detail.php?id=65504>

³ <https://www.eia.gov/todayinenergy/detail.php?id=41073>
<https://www.eia.gov/todayinenergy/detail.php?id=41073>

⁴ <https://unctad.org/publication/navigating-troubled-waters-impact-global-trade-disruption-shipping-routes-red-sea-black>
<https://unctad.org/publication/navigating-troubled-waters-impact-global-trade-disruption-shipping-routes-red-sea-black>

⁵ ⁷³ ⁹⁰ ⁹² <https://www.imf.org/en/blogs/articles/2024/03/07/red-sea-attacks-disrupt-global-trade>
<https://www.imf.org/en/blogs/articles/2024/03/07/red-sea-attacks-disrupt-global-trade>

⁶ ⁴⁹ ⁶⁵ ⁸⁴ <https://www.sabic.com/en/reports/integrated-report-2024/value-chain-impact/raw-materials>
<https://www.sabic.com/en/reports/integrated-report-2024/value-chain-impact/raw-materials>

⁷ ¹⁰ ¹⁷ ³⁹ ⁵⁹ ⁷⁵ ⁸¹ ⁸⁷ ⁹⁶ https://s23.q4cdn.com/981382065/files/doc_financials/2024/q4/DOW-INC-2024-10K-FINAL.pdf
https://s23.q4cdn.com/981382065/files/doc_financials/2024/q4/DOW-INC-2024-10K-FINAL.pdf

⁸ ⁵¹ ⁵² ⁵³ ⁶⁶ ⁶⁹ ⁷⁷ ⁸⁵ https://www.ineos.com/globalassets/investor-relations/public/annual-reports/ineos-group-holdings-s.a-audit-report-conso-31.12.2024-final_signed.pdf
https://www.ineos.com/globalassets/investor-relations/public/annual-reports/ineos-group-holdings-s.a-audit-report-conso-31.12.2024-final_signed.pdf

⁹ ³⁷ ⁴⁴ ⁴⁵ ⁴⁶ ⁴⁷ ⁶⁴ ⁷⁸ ⁸³ ⁹¹ https://report.basf.com/2024/en/_assets/downloads/combined-managements-report-basf-ar24.pdf
https://report.basf.com/2024/en/_assets/downloads/combined-managements-report-basf-ar24.pdf

11 40 41 42 60 61 62 63 76 82 88 93 <https://capedge.com/filing/1489393/0001489393-25-000008/LYB-10K-2024FY>

<https://capedge.com/filing/1489393/0001489393-25-000008/LYB-10K-2024FY>

12 13 55 70 86 94 97 <https://www.lobbyregister.bundestag.de/media/dd/bc/517969/Covestro-Annual-Report-2024.pdf>

<https://www.lobbyregister.bundestag.de/media/dd/bc/517969/Covestro-Annual-Report-2024.pdf>

15 56 <https://www.eia.gov/todayinenergy/detail.php?id=40152>

<https://www.eia.gov/todayinenergy/detail.php?id=40152>

16 <https://www.expandthehoustonschannel.com/project-overview/>

<https://www.expandthehoustonschannel.com/project-overview/>

18 <https://www.portofrotterdam.com/en/logistics/cargo/liquid-bulk/chemicals>

<https://www.portofrotterdam.com/en/logistics/cargo/liquid-bulk/chemicals>

19 <https://www.portofantwerpbruges.com/en/business/industry/chemicals>

<https://www.portofantwerpbruges.com/en/business/industry/chemicals>

20 <https://www.sabic.com/en/reports/integrated-report-2024/business-performance/chemicals>

<https://www.sabic.com/en/reports/integrated-report-2024/business-performance/chemicals>

21 <https://www.aramco.com/en/what-we-do/mega-projects/sadara-petrochemicals-facility>

<https://www.aramco.com/en/what-we-do/mega-projects/sadara-petrochemicals-facility>

22 <https://www.ecfr.gov/current/title-31/subtitle-B/chapter-V/part-560>

<https://www.ecfr.gov/current/title-31/subtitle-B/chapter-V/part-560>

23 <https://ofac.treasury.gov/sanctions-programs-and-country-information/iran-sanctions>

<https://ofac.treasury.gov/sanctions-programs-and-country-information/iran-sanctions>

24 <https://www.state.gov/releases/office-of-the-spokesperson/2025/10/sanctioning-entities-trading-in-iranian-petroleum-and-petrochemicals>

<https://www.state.gov/releases/office-of-the-spokesperson/2025/10/sanctioning-entities-trading-in-iranian-petroleum-and-petrochemicals>

25 57 79 80 <https://www.consilium.europa.eu/en/press/press-releases/2025/09/29/iran-sanctions-snapback-council-reimposes-restrictive-measures/>

<https://www.consilium.europa.eu/en/press/press-releases/2025/09/29/iran-sanctions-snapback-council-reimposes-restrictive-measures/>

26 <https://www.mayerbrown.com/en/insights/publications/2025/10/eu-reintroduces-sanctions-against-iran-following-un-snapback>

<https://www.mayerbrown.com/en/insights/publications/2025/10/eu-reintroduces-sanctions-against-iran-following-un-snapback>

27 <https://www.gov.uk/government/publications/iran-sanctions-guidance>

<https://www.gov.uk/government/publications/iran-sanctions-guidance>

29 98 <https://www.msc.com/en/newsroom/customer-advisories/2024/april/msc-aries-incident-update>

<https://www.msc.com/en/newsroom/customer-advisories/2024/april/msc-aries-incident-update>

30 <https://wwwcdn.imo.org/localresources/en/MediaCentre/Documents/Circular%20Letter%20No.4865%20-%20Communication%20From%20The%20Government%20Of%20Israel.pdf>

<https://wwwcdn.imo.org/localresources/en/MediaCentre/Documents/Circular%20Letter%20No.4865%20-%20Communication%20From%20The%20Government%20Of%20Israel.pdf>

31 <https://www.ukmto.org/>

<https://www.ukmto.org/>

33 <https://gard.no/en/circulars/notice-of-cancellation-for-war-risks/>

<https://gard.no/en/circulars/notice-of-cancellation-for-war-risks/>

34 100 <https://www.ft.com/content/503b9f75-f3a0-40b1-bc64-a9f2ffd7376f>

<https://www.ft.com/content/503b9f75-f3a0-40b1-bc64-a9f2ffd7376f>

35 <https://www.insurancejournal.com/news/international/2025/07/11/831268.htm>

<https://www.insurancejournal.com/news/international/2025/07/11/831268.htm>

36 <https://www.lloydslist.com/LL1147987/Suez-Canal-vessel-transits-plunge-34-in-seven-weeks-with-further-drop-on-cards>

<https://www.lloydslist.com/LL1147987/Suez-Canal-vessel-transits-plunge-34-in-seven-weeks-with-further-drop-on-cards>

38 101 <https://apnews.com/article/de5159966cde7de7b964b3c2c67eec07>

<https://apnews.com/article/de5159966cde7de7b964b3c2c67eec07>

43 <https://www.lyondellbasell.com/en/locations/africa-middle-east/saudi-arabia/>

<https://www.lyondellbasell.com/en/locations/africa-middle-east/saudi-arabia/>

48 50 74 95 <https://www.sabic.com/en/reports/integrated-report-2024/strategic-approach/risk-and-opportunities>

<https://www.sabic.com/en/reports/integrated-report-2024/strategic-approach/risk-and-opportunities>

54 68 89 <https://www.ineos.com/inch-magazine/articles/issue-16/welcome-to-the-kingdom/>

<https://www.ineos.com/inch-magazine/articles/issue-16/welcome-to-the-kingdom/>

67 99 https://unctad.org/system/files/non-official-document/cimem7-2024-ppt05ahmedelshazly_en.pdf

https://unctad.org/system/files/non-official-document/cimem7-2024-ppt05ahmedelshazly_en.pdf